EASY VISA PROJECT

Project and Course NamePost Graduate Program in Artificial Intelligence and Machine Learning

12/22/2025

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Executive Summary

- **The profile of the applicants for whose visa can be certified:**
- Education level - At least has a Bachelor's degree - Master's and doctorate are preferred.
- Job Experience - has job experience.
- Prevailing wage - has a high prevailing wage most likely yearly (The median prevailing wage of the employees for whom the visa got certified is around 72k.)
- Continent - it has been observed that applicants from Europe, Africa, and Asia have higher chances of visa certification.
- **The profile of the applicants for whom the visa status can be denied:**
- Education level - high school degree.
- Job Experience - Doesn't have any job experience.
- Prevailing wage and unit of wage - applicants with hourly units of wage (The median prevailing wage of the employees for whom the visa got certified is around 65k.)
- Continent - it has been observed that applicants from South America, North America, and Oceania have higher chances of visa applications getting

Business Problem Overview and Solution Approach

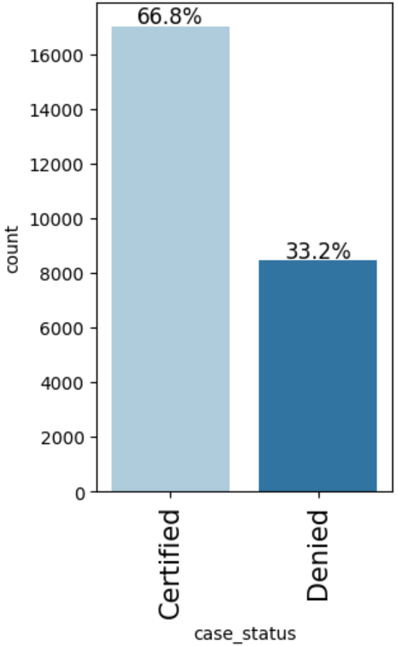
- OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in the area of intended employment.
- The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year. In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labor certifications which was a nine percent increase in the overall number of processed applications from the previous year.
- The task at hand to analyze the data provided to facilitate the process of visa approvals by building a machine learning model and to recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

Business Problem Overview and Solution Approach

- We want to predict whether the visa application will get certified or not using the information provided.
- We will use F1 Score as the metric for evaluation of the model because
    - Although since visa certification involves making important decisions about granting visas to employees, we need to be cautious about both false positives (approving visas for employees who shouldn't be eligible) and false negatives (rejecting visas for eligible employees). F1 score will help us to minimize both false positives and false negatives.
- We will use balanced class weights so that model focuses equally on both classes.
- We will build different models - DecisionTreeClassifier, RandomForestClassifier, AdaBoostClassifier, and GradientBoostingClassifierr.
- We will also perform hyperparameter tuning for these models and evaluate their performance using different metrics and confusion matrix.

EDA Results

Distribution of visa case statuses (certified vs. denied)



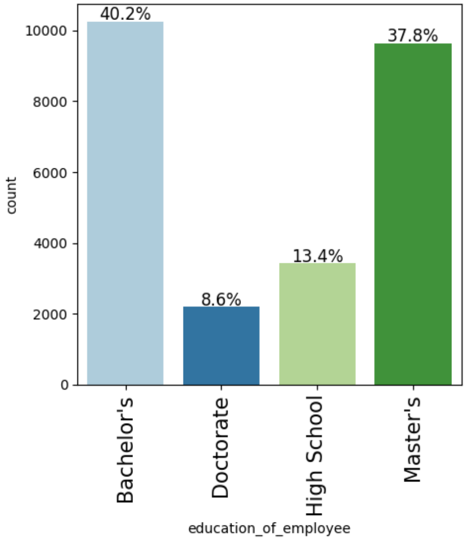
- The distribution on case status shows that it is not equally distributed.
- The most of the employees 17018 that were granted Visa which is 67% of the employed

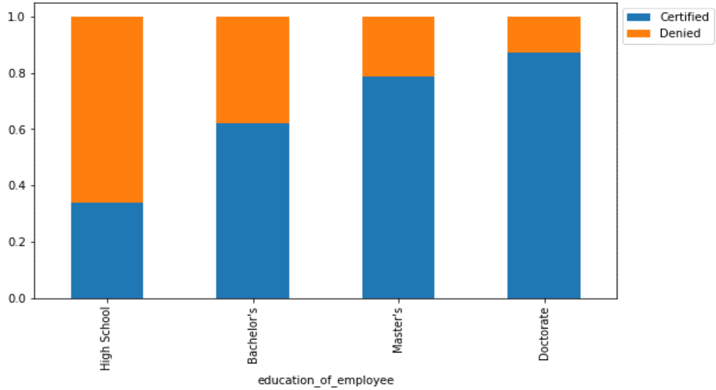
Link to Appendix slide on data background check

EDA Results

Education level impacting visa

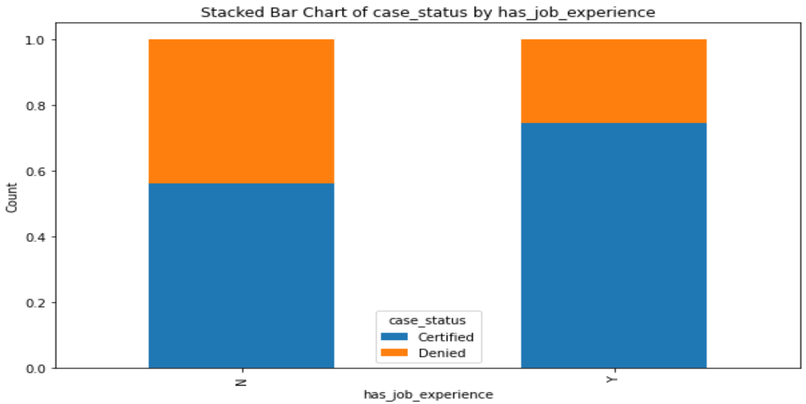
- 40.2% of the applicants have a bachelor's degree, followed by 37.8% having a master's degree.
- 8.6% of the applicants have a doctorate degree.
- Around 85% of applicants with Doctorate degree had their visa applications certified compared to Master's degree applicants that had 80% of the visa applications certified.
- Around 60% of the visa applications got certified for applicants with Bachelor's degrees.
- Applicants who do not have a degree and have graduated from high school are more likely to have their applications denied.

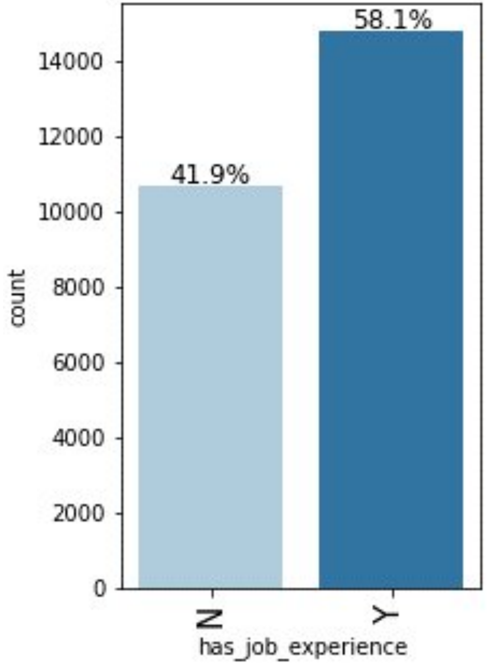




EDA Results

Difference in visa approval rates between employees with and without prior job experience



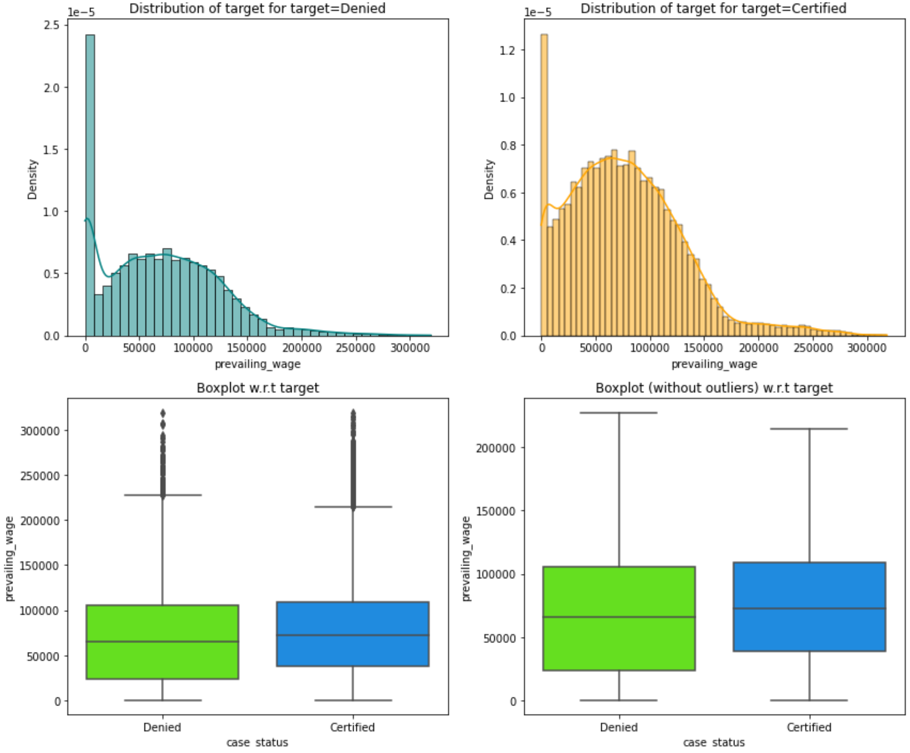


- THOSE THAT HAVE JOB EXPERIENCE HAVE A HIGHER CHANCE OF THEIR VISA BEING CERTIFIED WITH ABOUT 74%
- BUT ABOUT 56% OF PEOPLE WHO DIDNT HAVE A JOB EXPERIENCE ALSO GOT THEIR VISA CERTIFIED

EDA Results

Prevailing wage affect visa approval. Do higher wages lead to higher chances of approval

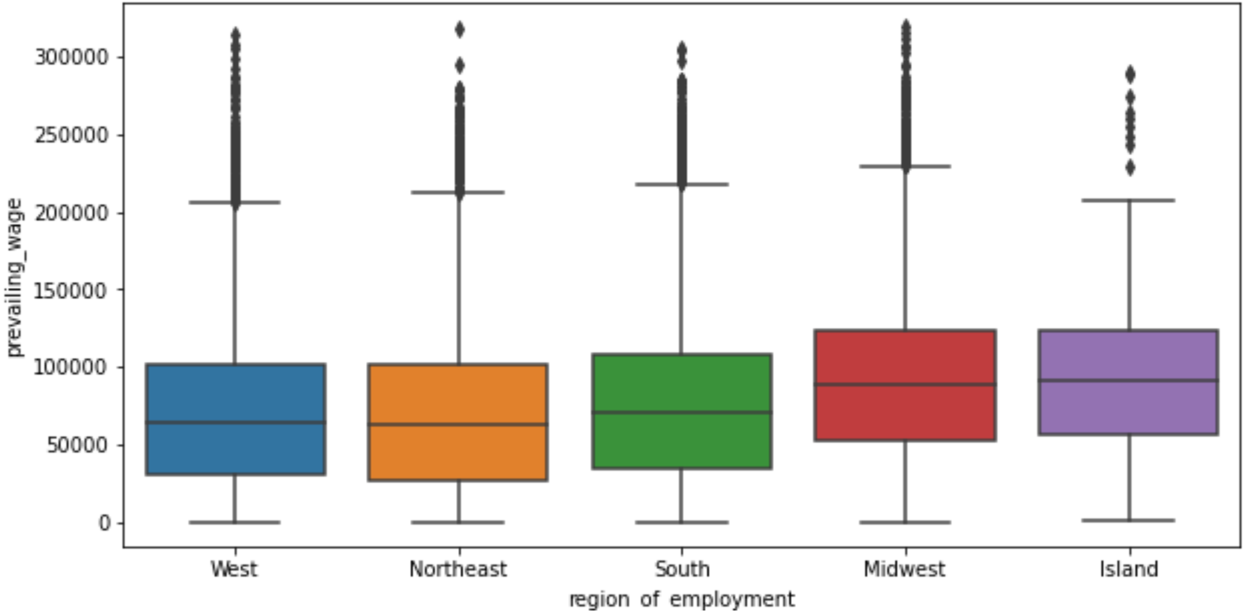
- The median prevailing wage for certified applications is slightly higher compared to denied applications.



EDA Results

Do certain regions in the US have higher visa approval rates compared to others

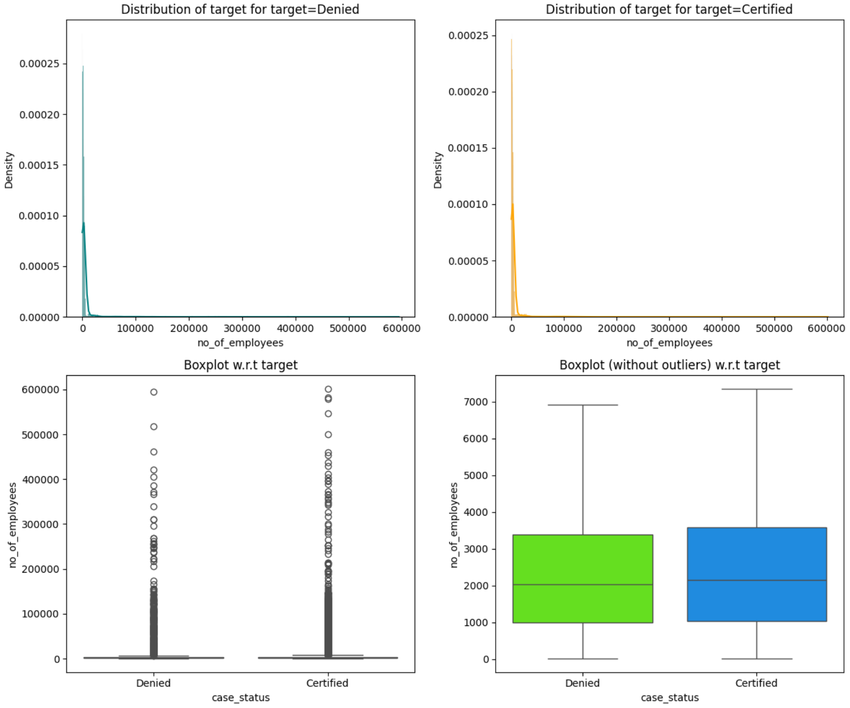
- Midwest and Island regions have slightly higher prevailing wages compared to other regions.
- The distribution of prevailing wage is similar across West, Northeast, and South regions.



EDA Results

Do larger companies have a higher approval rate?

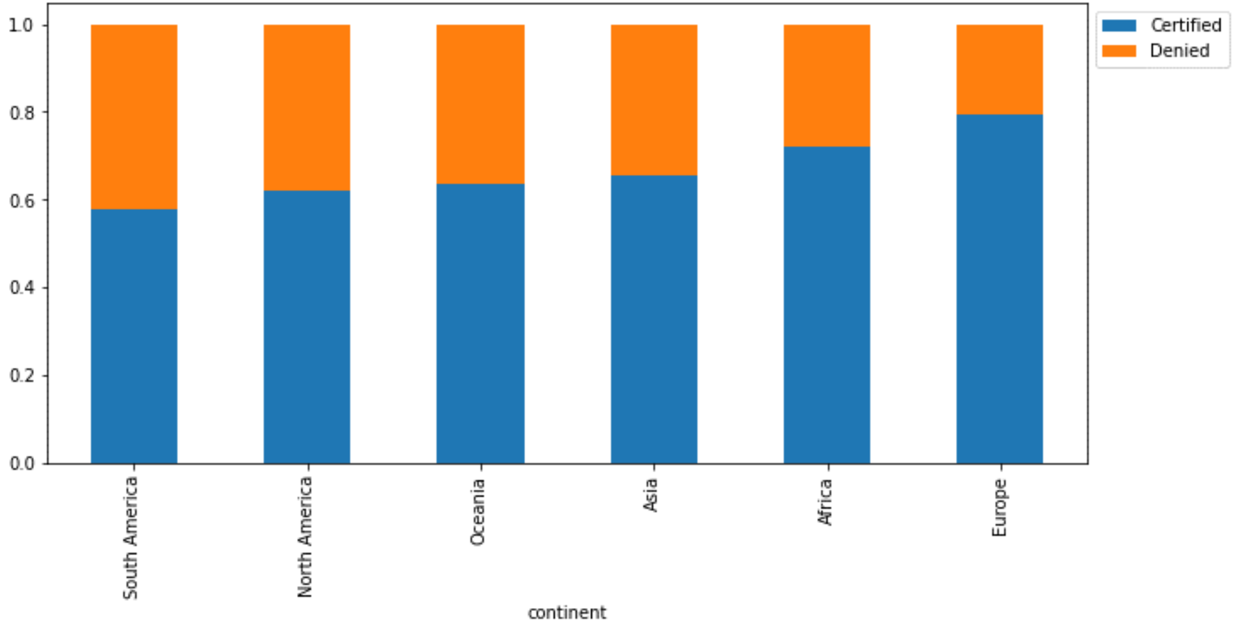
- Yes, larger companies tend to have a marginally higher approval rate — but company size is not a decisive factor.
- Visa approval is driven far more by role legitimacy, wages, and documentation quality than by employer size alone



EDA Results

Are visa approval rates different across various continents of employees? Which continent has the highest and lowest approval rates?

- Applications from Europe and Africa have a higher chance of getting certified.
- Around 80% of the applications from Europe are certified.
- Asia has the third-highest percentage (approx. 60%) of visa certification and has the highest number of applications.



EDA Results

Data Preprocessing

For each of the following steps, write down observations and describe the steps undertaken to preprocess the data

- Duplicate Value Check
- Missing Value Detection and Treatment (if needed with rationale)
- Outlier Detection and Treatment (if needed with rationale)
- Feature Engineering (if needed with rationale)
- Data Preparation for Modeling
    - Split the data into different sets
    - Encode the categorical variables

***Note**: You can use more than one slide if needed*

Duplicate Value Check

- There are no duplicate values

```
# checking for duplicate values
data.duplicated() ##  Complete the code to check for duplicate values
```

	0
0	False
1	False
2	False
3	False
4	False
...	...
25475	False
25476	False
25477	False
25478	False
25479	False

Missing Value

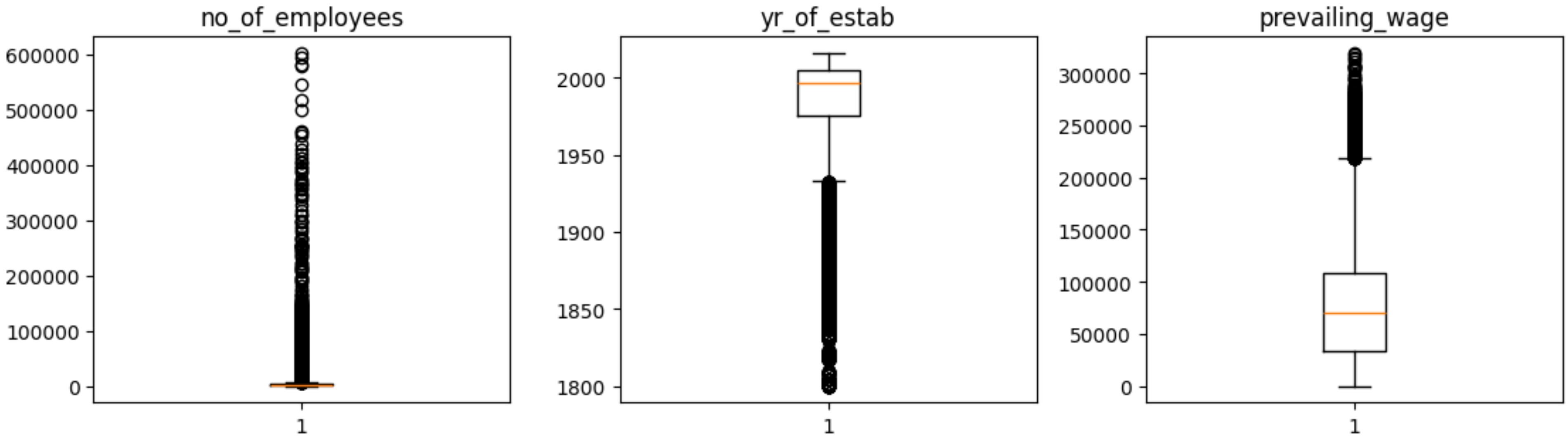
```
Null values case_id                  0
continent               0
education_of_employee   0
has_job_experience      0
requires_job_training   0
no_of_employees         0
yr_of_estab             0
region_of_employment    0
prevailing_wage         0
unit_of_wage            0
full_time_position      0
case_status             0
dtype: int64

NaN values case_id                   0
continent               0
education_of_employee   0
has_job_experience      0
requires_job_training   0
no_of_employees         0
yr_of_estab             0
region_of_employment    0
prevailing_wage         0
unit_of_wage            0
full_time_position      0
case_status             0
dtype: int64

There are no missing data points in the Personal Load dataset.
```

There are no missing values

Outlier detection and treatment



- There are outliers in the data.
- However, we will not treat them as they are proper values.

Model Performance Summary and Final Model Selection

- Compare the performance of the tuned models
- Overview of final ML model and its parameters
- Comment on the performance of the final model
- Summary of most important factors used by the ML model for prediction

***Note**: You can use more than one slide if needed*

Compare performance of tuned models

Based on the provided test performance comparison, the model with the highest overall F1-score is the
"Tuned Random Forest" classifier, with an F1-score of approximately 0.8209.

Training performance comparison:

Out[116]:

	Decision Tree	Tuned Decision Tree	Random Forest	Tuned Random Forest	Adaboost Classifier	Tuned Adaboost Classifier	Gradient Boost Classifier	Tuned Gradient Boost Classifier
Accuracy	1.0	0.712548	0.999944	0.769119	0.738226	0.719163	0.758802	0.764017
Recall	1.0	0.931923	0.999916	0.918660	0.887182	0.781415	0.883740	0.882649
Precision	1.0	0.720067	1.000000	0.776556	0.760688	0.794690	0.783042	0.789059
F1	1.0	0.812411	0.999958	0.841652	0.819080	0.787997	0.830349	0.833234

Testing performance comparison:

Out[114]:

	Decision Tree	Tuned Decision Tree	Random Forest	Tuned Random Forest	Adaboost Classifier	Tuned Adaboost Classifier	Gradient Boost Classifier	Tuned Gradient Boost Classifier
Accuracy	0.664835	0.706567	0.721088	0.738095	0.734301	0.716641	0.744767	0.743459
Recall	0.742801	0.930852	0.840744	0.898923	0.885015	0.781587	0.876004	0.871303
Precision	0.752232	0.715447	0.764926	0.755391	0.757799	0.791510	0.772366	0.773296
F1	0.747487	0.809058	0.801045	0.820930	0.816481	0.786517	0.820927	0.819379

Model Performance Comparison

Training performance comparison:

	Decision Tree	Tuned Decision Tree	Bagging Classifier	Tuned Bagging Classifier	Random Forest	Tuned Random Forest	Adaboost Classifier	Tuned Adaboost Classifier	Gradient Boost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier	XGBoost Classifier Tuned	Stacking Classifier
Accuracy	1.0	0.712548	0.985198	0.996187	1.0	0.769119	0.738226	0.718995	0.758802	0.764017	0.838753	0.767493	0.769399
Recall	1.0	0.931923	0.985982	0.999916	1.0	0.918660	0.887182	0.781247	0.883740	0.882649	0.931419	0.882565	0.892135
Precision	1.0	0.720067	0.991810	0.994407	1.0	0.776556	0.760688	0.794587	0.783042	0.789059	0.843482	0.792791	0.789834
F1	1.0	0.812411	0.988887	0.997154	1.0	0.841652	0.819080	0.787861	0.830349	0.833234	0.885272	0.835273	0.837873

Tesing performance comparison:

	Decision Tree	Tuned Decision Tree	Bagging Classifier	Tuned Bagging Classifier	Random Forest	Tuned Random Forest	Adaboost Classifier	Tuned Adaboost Classifier	Gradient Boost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier	XGBoost Classifier Tuned	Stacking Classifier
Accuracy	0.652538	0.706567	0.691523	0.724228	0.727368	0.738095	0.734301	0.716510	0.744767	0.743459	0.733255	0.746075	0.743721
Recall	0.728306	0.930852	0.764153	0.895397	0.847209	0.898923	0.885015	0.781391	0.876004	0.871303	0.860725	0.870715	0.878159
Precision	0.745538	0.715447	0.771711	0.743857	0.768343	0.755391	0.757799	0.791468	0.772366	0.773296	0.767913	0.776284	0.770275
F1	0.736821	0.809058	0.767913	0.812622	0.805851	0.820930	0.816481	0.786397	0.820927	0.819379	0.811675	0.820792	0.820686

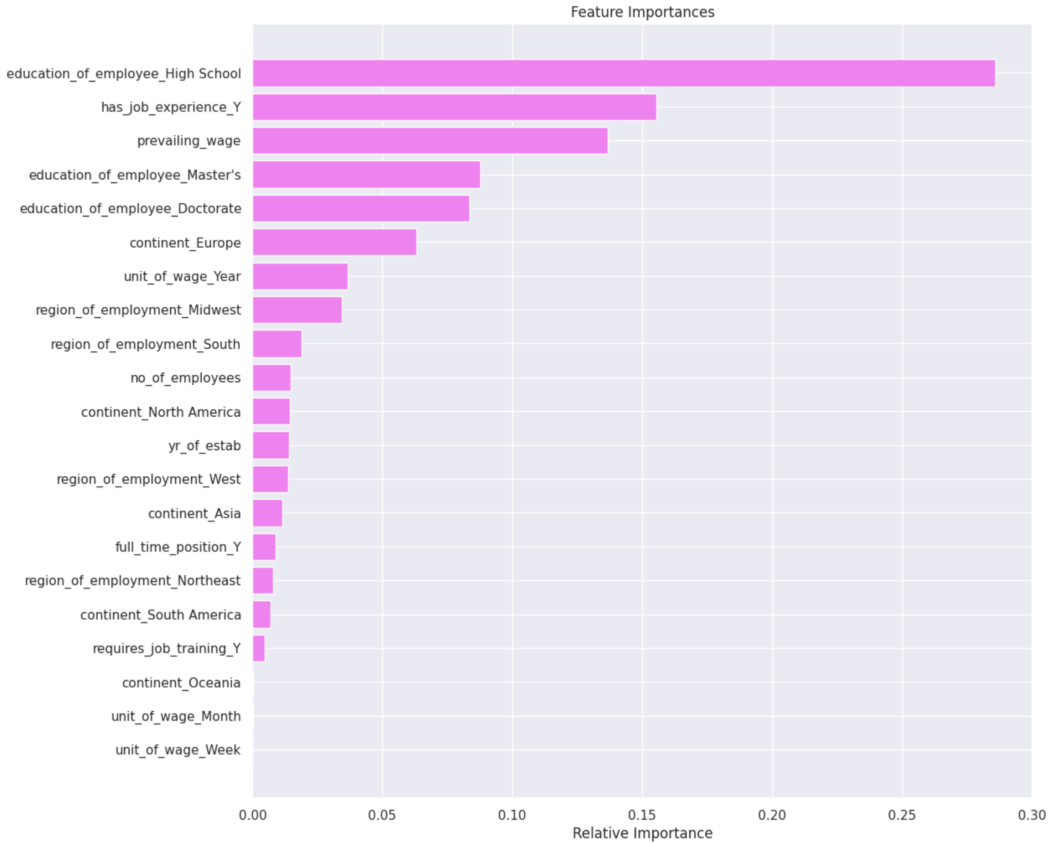
Observations on Performance Comparison

- Decision Tree overfitted the training data before tunning and performed poorly on the test data.
- Tunned Decision tree performed well on both the training and test data
- Bagging claassifier overfitted the training data before and after tunning on the training data but performed poorly on the test data.
- Random forest overfitted the training data before tuning
- Tuned random forest performed well on training and test data
- Adaboost model performed well before and after tunning on both training and test data.
- Gradient boost and XGBoost model performed well before and after tuning on both training and test data
- Stacking model also performed well on both training and test data.
- Tuned decision tree gave the highest recall on test data 0.930852

Actionable Insights and Recommendations

- We have been able to build a predictive model: a) that Office of Foreign Labor Certification (OFLC) can deploy this model to Facilitate the process of visa approvals. b) This will help OFLC have an idea of a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status. c) based on which OFLC can take appropriate actions to build better Visa approval policies.
- Factors that leads to Visa certification - high school certification (education_of employee_high_school), having job experience (has_job_experience), prevailing wage and Employee with masters certification (education_of employee_masters)
- high school certification : You need to at least attend high school certification in order to get a Visa. This is because employers in US needs employees that has some level of talent and education
- Prevailing wage: Having less prevailing wage help the employer same more money and gain more.
- Employee with masters certification: Those with masters certification seems to add more values to the workforce of the employers company and help increase profit becaue of more exposure in their field of study.

Feature Importance



Employee with high school certification (education_of employee_high_school) is the most important feature for prediction followed by having job experience (has_job_experience), prevailing wage and Employee with masters certification (education_of employee_masters)

APPENDIX

Great Learning

Happy Learning !